Quantum Behavior as Wave Interference from Higher-Dimensional Scalar Fields

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Abstract

We propose a geometric framework in which quantum-like behavior arises from deterministic resonance between classical scalar fields defined on a recursively modulated higher-dimensional manifold. The observable universe is modeled as a three-dimensional spatial hypersurface embedded within a six-dimensional flat Lorentzian space. Recursive curvature, encoded via smooth functions in the extra dimensions, modulates the phase and amplitude of a classical ambient field evaluated on this hypersurface. We show that the resulting field structure exhibits interference, localization, and modulation effects commonly associated with quantum phenomena, without invoking stochastic processes or probabilistic postulates. A confined scalar field sourcing and responding to the ambient field yields an effective Schrödinger-like equation with a dynamically induced potential, suggesting that quantum behavior can emerge from classical field interaction under recursive geometric alignment. Unlike holographic or brane-world models, our framework is not based on boundary duality but on recursive curvature as an organizing principle. The resulting modulation reproduces nodal structures, localization patterns, and decoherence-like suppression as deterministic consequences of classical wave resonance. This model provides a viable testbed for reinterpreting quantum interference as emergent from recursive geometry, with implications for fields, gravity, and fundamental ontology.

Keywords: recursive modulation, deterministic interference, classical field resonance, geometric coupling, analog quantum simulation, envelope suppression

1 Introduction

The emergence of quantum behavior from classical systems has long intrigued physicists, particularly in higher-dimensional frameworks that aim to unify geometry with field dynamics. While traditional approaches such as Kaluza–Klein theory and brane-world models [12, 13, 16, 17] emphasize gravitational unification or field localization, the present work explores how interference, localization, and measurement-like features can arise from deterministic resonance between scalar fields layered across recursively overlapping dimensions.

We model the observable universe as a three-dimensional spatial submanifold recursively stabilized within a six-dimensional flat Lorentzian manifold. Here, the "fourth dimension" serves not as time, but as a structural reference surface that organizes the recursive coupling between the confined 3D region and the ambient 6D field environment, almost as a dimensional atmosphere.

Within this geometry, both the ambient scalar field and the confined matter field exhibit wave-like behavior. Their physical overlap produces structured interference as an expected result of deterministic wave-on-wave resonance, not as a probabilistic anomaly. These interactions reproduce quantum interference and localization with high fidelity [4].

It is important to emphasize that in this model, what appears as quantum behavior is not due to spacetime curvature, but to phase alignment between coexisting classical fields. Curvature acts only to mediate the recursive dimensional layering, but it does not induce quantization.

We therefore aim to model a self-consistent system in which the confined field both sources and responds to the ambient field via Green's function dynamics. In the semiclassical limit, this appears to yield an effective Schrödinger-like equation with a modulation-induced potential, suggesting that quantum phenomena may, and perhaps can, emerge from classical field interaction across recursive geometric structures [5, 8, 23].

The model generalizes naturally to fermions, gauge fields, and curvature modes, offering a deterministic, testable framework for quantum behavior.

Although this approach may echo some terminology used in holographic models ("projection" or "embedding"), it is not based on duality or information transference across boundaries. Observable structure here arises from recursive geometric resonance, governed by spin, curvature, and parity-filtered dimensional descent, as will be explored in later works [18].

2 Geometric Recursive Framework

Let $(\mathcal{M}_6, \eta_{AB})$ be a six-dimensional Lorentzian manifold equipped with a flat metric

$$\eta_{AB} = \mathrm{diag}(-1, +1, +1, +1, +1, +1),$$

where indices $A, B \in \{0, 1, 2, 3, 4, 5\}$. We define a recursively stabilized four-dimensional submanifold $\Sigma_4 \subset \mathcal{M}_6$, which represents the observable spacetime.

Coordinates on $\mathcal{M}_6$ are denoted by $x^A = (x^\mu, x^4, x^5)$, where Greek indices $\mu, \nu \in \{0,1,2,3\}$ correspond to intrinsic coordinates on Σ_4.

The structural recursion defining Σ_4 is governed by two smooth scalar functions:

$$x^4 = f_1(x^\mu), \qquad x^5 = f_2(x^\mu),$$

so that a point on the recursive layer is given in ambient coordinates by

$$x^A(x^\mu) = \big(x^\mu, f_1(x^\mu), f_2(x^\mu)\big).$$

The geometry induced on Σ_4 is determined by the internal recursion of the ambient metric through this structure. The tangent vectors to the recursive layer are given by

$$e^A_\mu = \frac{\partial x^A}{\partial x^\mu} = \left(\delta^A_\mu, \partial_\mu f_1, \partial_\mu f_2\right),$$

which define a basis for the tangent bundle $T\Sigma_4$. The induced metric is computed as

$$g_{\mu\nu} = \eta_{AB}\, e^A_\mu\, e^B_\nu = \eta_{\mu\nu} + \partial_\mu f_1\, \partial_\nu f_1 + \partial_\mu f_2\, \partial_\nu f_2.$$

To describe the extrinsic geometry, we introduce two normal vector fields $n^{(1)A}$ and $n^{(2)A}$ satisfying

$$\eta_{AB}\, n^{(i)A}\, e^B_\mu = 0, \qquad \eta_{AB}\, n^{(i)A}\, n^{(j)B} = \delta^{ij},$$

for $i, j \in \{1, 2\}$. An explicit orthonormal frame can be constructed as

$$n^{(1)A} = \frac{1}{\sqrt{1 + \partial^\rho f_1\, \partial_\rho f_1}} \left(-\partial^\mu f_1, 1, 0\right), \qquad n^{(2)A} = \frac{1}{\sqrt{1 + \partial^\rho f_2\, \partial_\rho f_2}} \left(-\partial^\mu f_2, 0, 1\right),$$

where contractions such as $\partial^\rho f_i\, \partial_\rho f_i$ are taken with respect to $\eta^{\mu\nu}$, the inverse of the ambient metric restricted to the first four coordinates.

The extrinsic curvature tensors $K^{(i)}_{\mu\nu}$ associated with each normal direction are defined as

$$K^{(i)}_{\mu\nu} = -n^{(i)}_B\, \frac{\partial^2 x^B}{\partial x^\mu \partial x^\nu},$$

yielding

$$K^{(1)}_{\mu\nu} = -\frac{\partial_\mu \partial_\nu f_1}{\sqrt{1 + \partial^\rho f_1\, \partial_\rho f_1}}, \qquad K^{(2)}_{\mu\nu} = -\frac{\partial_\mu \partial_\nu f_2}{\sqrt{1 + \partial^\rho f_2\, \partial_\rho f_2}}.$$

The full extrinsic curvature vector $K^A_{\mu\nu}$ takes values in the normal bundle and is given by

$$K^A_{\mu\nu} = K^{(1)}_{\mu\nu}\, n^{(1)A} + K^{(2)}_{\mu\nu}\, n^{(2)A}.$$

The invariant volume element induced on Σ_4 by $g_{\mu\nu}$ is

$$\sqrt{-\det g_{\mu\nu}}\, \mathrm{d}^4 x.$$

We now introduce a real scalar field $\Phi(x^A)$ defined on $\mathcal{M}_6$, and consider its evaluation on the recursive layer:

$$\phi(x^\mu) = \Phi\big(x^\mu, f_1(x^\mu), f_2(x^\mu)\big).$$

The derivatives of ϕ can be expressed via the chain rule as

$$\partial_\mu \phi = \frac{\partial \Phi}{\partial x^A} \frac{\partial x^A}{\partial x^\mu} = e^A_\mu \, \partial_A \Phi.$$

To compute the Laplacian of ϕ with respect to the induced geometry, we define the d'Alembert operator associated with $g_{\mu\nu}$,

$$\Box_4 \phi = g^{\mu\nu} \nabla_\mu \nabla_\nu \phi = \frac{1}{\sqrt{-\det g}} \, \partial_\mu \left(\sqrt{-\det g} \, g^{\mu\nu} \, \partial_\nu \phi \right).$$

Using the identity $\partial_\mu \phi = e^A_\mu \, \partial_A \Phi$, we obtain

$$g^{\mu\nu} \, \partial_\mu \phi \, \partial_\nu \phi = g^{\mu\nu} \, e^A_\mu \, e^B_\nu \, \partial_A \Phi \, \partial_B \Phi = P^{AB} \, \partial_A \Phi \, \partial_B \Phi,$$

where

$$P^{AB} = g^{\mu\nu} \, e^A_\mu \, e^B_\nu$$

acts as a recursion-aligned projection operator from ambient tensors to their tangential components on Σ_4. The wave operator acting on the recursively evaluated field can therefore be written as

$$\Box_4 \phi = P^{AB} \, \nabla_A \nabla_B \Phi + (\text{extrinsic curvature terms}).$$

In the limit of weak curvature—when $|\partial_\mu f_i| \ll 1$—we may expand the kinetic term:

$$\begin{aligned} \eta^{\mu\nu} \, \partial_\mu \phi \, \partial_\nu \phi &= (\partial^\mu \Phi + \partial_4 \Phi \, \partial^\mu f_1 + \partial_5 \Phi \, \partial^\mu f_2) \, (\partial_\mu \Phi + \partial_4 \Phi \, \partial_\mu f_1 + \partial_5 \Phi \, \partial_\mu f_2) \\ &= \partial^\mu \Phi \, \partial_\mu \Phi + 2 \partial^\mu \Phi \, (\partial_4 \Phi \, \partial_\mu f_1 + \partial_5 \Phi \, \partial_\mu f_2) + \cdots , \end{aligned}$$

where again the ellipsis denotes higher-order curvature corrections. This expansion makes clear the coupling between ambient dynamics and recursive geometric modulation. While in this section we consider the case where $\Phi(x^A)$ is a prescribed harmonic function in the ambient space, the more complete model developed in later sections allows Φ to be dynamically sourced by the confined matter field $\psi(x^\mu)$. The recursion structure derived here remains valid in either case, and provides the geometric foundation for the modulation effects central to the resonance-based mechanism explored in the remainder of the paper [14, 18, 21].

3 Bulk Scalar Field and Recursive Interference

3.1 Bulk Field Action and Equations of Motion

We consider a real scalar field $\Phi(x^A)$ defined on the six-dimensional Lorentzian manifold $(\mathcal{M}_6, \eta_{AB})$, where the ambient metric is flat with signature $(-+++++)$. The dynamics of Φ are governed by the standard massless action:

$$S_\Phi = -\frac{1}{2}\int_{\mathcal{M}_6} \mathrm{d}^6x\,\sqrt{-\eta}\,\eta^{AB}\partial_A\Phi\,\partial_B\Phi,$$

where $\eta = \det(\eta_{AB}) = -1$, so $\sqrt{-\eta} = 1$. Variation with respect to Φ yields the Euler–Lagrange equation:

$$\frac{\delta S_\Phi}{\delta\Phi} = \partial_A\left(\eta^{AB}\partial_B\Phi\right) = 0 \quad \Rightarrow \quad \Box_6\Phi = \eta^{AB}\partial_A\partial_B\Phi = 0.$$

3.2 Mode Structure and On-Shell Condition

We consider harmonic plane wave solutions of the form:

$$\Phi(x^A) = \cos\left(k_\mu x^\mu + \omega_4 x^4 + \omega_5 x^5\right),$$

where k_μ is the four-dimensional wavevector, and ω_4, ω_5 are spatial frequencies in the extra dimensions. Substituting into the wave equation gives the on-shell condition:

$$-k^\mu k_\mu + \omega_4^2 + \omega_5^2 = 0.$$

This relation ensures that the wave propagates at the speed of light in the ambient 6D space. The field $\Phi(x^A)$ represents a classical oscillatory mode in the higher-dimensional structure.

3.3 Recursive Evaluation and Modulated Field

The physical universe is modeled as a four-dimensional hypersurface Σ_4, structurally embedded in the six-dimensional flat manifold $\mathcal{M}_6$ via recursive functions

$$x^4 = f_1(x^\mu), \quad x^5 = f_2(x^\mu).$$

While Σ_4 is formally a four-dimensional submanifold, its intrinsic spatial geometry is that of a closed three-sphere S^3, with curvature encoded recursively in the embedding functions. This perspective allows Σ_4 to appear locally as a curved hypersurface, while globally forming a structure diffeomorphic to $S^3\times\mathbb{R}$ when viewed from $\mathcal{M}_6$—analogous to how Earth's atmosphere appears locally flat but globally spherical.

A point on the recursive layer is represented by:

$$x^A(x^\mu) = (x^\mu, f_1(x^\mu), f_2(x^\mu)).$$

The scalar field observable on Σ_4 is evaluated recursively as:

$$\phi(x^\mu) = \Phi(x^A(x^\mu)) = \Phi(x^\mu, f_1(x^\mu), f_2(x^\mu)).$$

The derivatives of ϕ follow from the chain rule:

$$\partial_\mu \phi = \partial_\mu \Phi + \partial_4 \Phi \cdot \partial_\mu f_1 + \partial_5 \Phi \cdot \partial_\mu f_2 = e^A_\mu \, \partial_A \Phi,$$

where the tangent vectors $e^A_\mu = \partial x^A / \partial x^\mu$ define the recursive curvature geometry. This establishes the alignment of ambient derivatives with the local structure.

It is important to clarify that while the hypersurface Σ_4 has four intrinsic dimensions (three spatial and one temporal), the emergence of interference and modulation phenomena in this model is not caused by spacetime curvature alone, but by the recursive alignment of oscillating ambient fields with the geometry of the hypersurface. Both the confined matter field $\psi(x^\mu)$ and the ambient bulk field $\Phi(x^A)$ are classical wave solutions. They inherently exhibit temporal and spatial oscillation. The ambient field propagates freely in flat six-dimensional space, while the confined field evolves on the recursively curved 4D surface. When the ambient field is evaluated on the hypersurface, the recursive curvature of the three-dimensional spatial submanifold modulates the phase and amplitude of this interaction. Thus, the resulting interference patterns and structured modulation arise from the resonance between coexisting wave fields, mediated by recursive geometry. The curvature does not generate the structure, but rather refracts and aligns it [6, 9].

The observed quantum-like behavior in this system is therefore a consequence of deterministic wave-on-wave resonance under recursive curvature, not an effect of curvature alone.

This model is not holographic in the AdS/CFT or boundary duality sense. The recursion described here does not encode bulk dynamics onto a lower-dimensional boundary, but instead produces emergent structure through internal geometric layering and resonance across dimensional strata [1].

3.4 Covariant Definition of the Modulation Scalar

The recursion functions may carry periodic dependence on a scalar function $S(x^\mu)$ that modulates the geometry. To preserve four-dimensional Lorentz covariance, we define:

$$S(x^\mu) = k_\mu x^\mu = -\omega x^0 + k^i x^i,$$

where k_μ is a fixed reference wavevector and summation over spatial indices $i = 1, 2, 3$ is implied. Under Lorentz transformations,

$$S'(x') = k'_\mu x'^\mu = k_\mu x^\mu = S(x),$$

ensuring $S(x^\mu)$ is a true scalar. In special cases, such as working in a preferred cosmological or lab frame, this reduces to the simpler form $S = x^0 + x^1 + x^2 + x^3$, but we adopt the covariant definition throughout.

3.5 Flat vs. Rippled Geometry: Interference Emergence

If the hypersurface is flat, i.e.,

$$f_1(x^\mu) = 0, \quad f_2(x^\mu) = 0,$$

then the recursive field reduces to:

$$\phi_{\text{flat}}(x^\mu) = \Phi(x^\mu, 0, 0) = \cos(k_\mu x^\mu),$$

which is a smooth plane wave on the hypersurface.

To introduce modulation, we consider a rippled recursive geometry:

$$f_1(x^\mu) = \epsilon \sin(\lambda S(x^\mu)), \quad f_2(x^\mu) = \epsilon \cos(\lambda S(x^\mu)),$$

with small amplitude ϵ and modulation frequency λ. Substituting into the bulk field gives:

$$\begin{aligned}\phi(x^\mu) &= \Phi\left(x^\mu, f_1(x^\mu), f_2(x^\mu)\right) \\ &= \cos\left(k_\mu x^\mu + \omega_4 \epsilon \sin(\lambda S) + \omega_5 \epsilon \cos(\lambda S)\right).\end{aligned}$$

Using the identity

$$\omega_4 \sin(\lambda S) + \omega_5 \cos(\lambda S) = A \sin(\lambda S + \theta),$$

with

$$A = \sqrt{\omega_4^2 + \omega_5^2}, \quad \theta = \tan^{-1}\left(\frac{\omega_5}{\omega_4}\right),$$

we obtain the modulated form:

$$\phi_{\text{rippled}}(x^\mu) = \cos\left(k_\mu x^\mu + \epsilon A \sin(\lambda S(x^\mu) + \theta)\right).$$

This is a phase-modulated wave, exhibiting a nonlinear envelope structure determined by recursive curvature. Even though the bulk field Φ is a simple harmonic plane wave, the observable field ϕ acquires structured modulation due to the recursive geometry [4, 9].

Although the interference visually resembles holographic modulation, this framework operates entirely within a recursive geometric system. The observed effects emerge from structural wave coupling and dimensional resonance, not from any dual boundary representation.

3.6 Interpretation

This expansion makes attempts to make explicit how recursive geometry affects the local structure of confined fields. However, the quantum-like modulation observed in this model does not arise from projection distortion or curvature alone. Instead, the core mechanism is the deterministic interaction between the ambient scalar field

$\Phi(x^A)$, propagating in 6D, and the confined field $\psi(x^\mu)$, which both sources and responds to it. The recursive structure derived here provides a consistent mathematical framework for describing how fields defined in different dimensional layers coexist and exchange information. These dynamics ultimately give rise to the interference and modulation patterns explored in subsequent sections as emergent resonance behavior within shared recursive structure.

From within the hypersurface, the field $\psi(x^\mu)$ appears to exhibit quantum behavior. From the higher-dimensional view, the same structure is a classical recursive field oscillation. These are not separate phenomena but two expressions of a single underlying wave system: curvature and field, entangled through recursive geometry. In this model, what appears as quantum interference is the interior expression of structural resonance, and what appears as fluctuating geometry is the outward imprint of wave coherence. They are, in essence, the same object, seen from within and from above.

4 Mutual Resonance Between Confined Matter Fields and Ambient Geometry

We now construct a self-consistent framework in which a real scalar field confined to a three-dimensional recursive hypersurface interacts bidirectionally with a higher-dimensional ambient field. The core mechanism is resonant. The confined matter field acts as a localized source for the ambient scalar field, which in turn modulates the matter dynamics via recursive geometric alignment. This coupling seems to induce structured interference phenomena without invoking stochasticity or quantum postulates, and arises entirely from classical field equations within a curvature-regulated recursion [2, 8].

4.1 Ambient Field with Recursively Localized Source

Let $\Phi(x^A)$ be a real scalar field defined on a six-dimensional flat Lorentzian manifold $(\mathcal{M}_6, \eta_{AB})$, where $x^A = (x^\mu, x^4, x^5)$, with $\mu \in \{0,1,2,3\}$. The field Φ satisfies the inhomogeneous wave equation

$$\Box_6 \Phi(x^A) = J(x^A), \tag{1}$$

where $\Box_6 = \eta^{AB}\partial_A\partial_B$, and $J(x^A)$ is a source localized on a smooth recursive hypersurface Σ_4. The recursion is defined by two functions $f_1(x^\mu)$ and $f_2(x^\mu)$ specifying the transverse coordinates $x^4 = f_1(x^\mu)$, $x^5 = f_2(x^\mu)$. The source is then written as

$$J(x^A) = \delta\big(x^4 - f_1(x^\mu)\big)\,\delta\big(x^5 - f_2(x^\mu)\big)\,\mathcal{J}(x^\mu), \tag{2}$$

where $\mathcal{J}(x^\mu)$ is a scalar source current defined along the recursive structure. We take

$$\mathcal{J}(x^\mu) = \alpha\,\psi(x^\mu), \tag{3}$$

with $\alpha \in \mathbb{R}$, and $\psi(x^\mu)$ a real scalar matter field confined to Σ_4.

4.2 Solution via Ambient Green's Function

Let $G(x^A; x'^A)$ denote the retarded Green's function for the ambient operator, satisfying

$$\Box_6 G(x^A; x'^A) = \delta^{(6)}(x^A - x'^A). \tag{4}$$

The solution to the sourced equation is given by

$$\Phi(x^A) = \int_{\mathbb{R}^4} \alpha\, \psi(x'^\mu)\, G\Big(x^A;\, x'^\mu, f_1(x'^\mu), f_2(x'^\mu)\Big)\, \mathrm{d}^4x'. \tag{5}$$

This expresses the ambient field at arbitrary points as a convolution over the recursive layer with a source distributed across its curvature-aligned structure [14, 21].

The integral expression for $\Phi(x^A)$ in Eq. (5) assumes that the Green's function $G(x^A; x'^A)$ is well-defined and that the convolution over the recursive geometry yields finite results. In general, the Green's function for the 6D d'Alembertian is supported on the ambient light cone:

$$\operatorname{supp}\big(G(x^A; x'^A)\big) \subseteq \{(x^A, x'^A) \mid (x^A - x'^A)^2 = 0\}.$$

For smooth recursion and localized source fields $\psi(x^\mu)$ of compact support, the resulting integrals are expected to converge due to the delta-function localization in x^4 and x^5 (Eq. (2)). However, future work may need to analyze the infrared and ultraviolet behavior of G for recursion schemes with large extrinsic curvature gradients.

At spatial infinity, we impose outgoing (radiative) boundary conditions in the ambient space, in accordance with causal propagation and the canonical construction of retarded Green's functions.

4.3 Recursive Evaluation and Induced Interaction

The induced field $\phi(x^\mu)$, evaluated along the recursive layer, is given by:

$$\phi(x^\mu) = \Phi\big(x^\mu, f_1(x^\mu), f_2(x^\mu)\big). \tag{6}$$

Substituting the integral representation, we obtain

$$\phi(x^\mu) = \alpha \int_{\mathbb{R}^4} \psi(x'^\mu)\, G\Big(x^\mu, f_1(x^\mu), f_2(x^\mu);\, x'^\mu, f_1(x'^\mu), f_2(x'^\mu)\Big)\, \mathrm{d}^4x'. \tag{7}$$

The field ϕ is thus a nonlocal functional of ψ, with kernel determined jointly by the ambient Green's function and the recursive curvature of the hypersurface.

4.4 Self-Consistent Matter Field Dynamics

The matter field $\psi(x^\mu)$ evolves according to a curved-spacetime Klein–Gordon-type equation with interaction term sourced by ϕ:

$$\big(\Box_4 + m^2 + 2g\, \phi(x^\mu)\big)\, \psi(x^\mu) = 0, \tag{8}$$

where $m > 0$ is the rest mass and $g \in \mathbb{R}$ a coupling constant. Substituting the expression for ϕ, we obtain a nonlinear, nonlocal equation of motion:

$$\left(\Box_4 + m^2\right) \psi(x^\mu) + 2\alpha g \int_{\mathbb{R}^4} G_{\text{eff}}(x^\mu, x'^\mu)\, \psi(x'^\mu)\, \mathrm{d}^4x' = 0, \tag{9}$$

where the effective kernel is given by

$$G_{\text{eff}}(x^\mu, x'^\mu) = G\Big(x^\mu, f_1(x^\mu), f_2(x^\mu);\; x'^\mu, f_1(x'^\mu), f_2(x'^\mu)\Big). \tag{10}$$

4.5 Interpretation

This formulation describes a fully coupled classical system in which the confined matter field emits ambient radiation into six-dimensional space and responds dynamically to the returning wave structure. The interference patterns and localization effects observed in the model are not caused by the geometry of the hypersurface alone, but by the deterministic interaction between coexisting wave systems in recursively stabilized geometric strata. The confined field $\psi(x^\mu)$ sources the ambient scalar field $\Phi(x^A)$, which propagates in 6D and reflects back into the confined domain. The result appears to be a real-space resonance between ambient and confined fields, acting as a recursive wave interaction that produces structured amplitude modulation. The kernel G_{eff} captures this feedback precisely, and requires no stochastic terms, collapse postulates, or quantum axioms. Quantum-like behavior thus, in this model, emerges naturally as an interference phenomenon arising from recursive classical geometry [8, 23].

Effective Action and Semiclassical Limit

We now examine the effective dynamics of a real scalar matter field $\psi(x^\mu)$ confined to the recursive hypersurface Σ_4, taking into account the modulation induced by its deterministic interaction with the ambient scalar field $\Phi(x^A)$. As described previously, the recursively evaluated field $\phi(x^\mu) = \Phi\big(x^\mu, f_1(x^\mu), f_2(x^\mu)\big)$ is a nonlocal functional of $\psi(x^\mu)$, determined by the ambient Green's function and the structural resonance across recursive geometric strata. While the recursive geometry organizes this alignment, the underlying modulation arises from active resonance between scalar fields occupying distinct layers of a unified recursive system.

To study localized, low-energy behavior, we adopt a semiclassical approximation in which $\phi(x^\mu)$ is treated as a background potential varying on length and time scales much larger than the rapid phase oscillations of ψ [3, 23]. This separation of scales permits a consistent truncation of the coupled system, allowing ϕ to be held fixed when deriving the non-relativistic dynamics of ψ.

The effective action for ψ on Σ_4 is then given by

$$S[\psi] = \int_{\Sigma_4} \left(-\frac{1}{2} g^{\mu\nu} \partial_\mu \psi\, \partial_\nu \psi - \frac{1}{2} m^2 \psi^2 - g\, \phi(x^\mu)\, \psi^2 \right) \sqrt{-\det g}\, \mathrm{d}^4x, \tag{11}$$

where $m > 0$ is the rest mass of the matter field and $g > 0$ is a coupling constant. Variation yields the classical equation of motion

$$\frac{1}{\sqrt{-\det g}}\,\partial_\mu\left(\sqrt{-\det g}\,g^{\mu\nu}\partial_\nu\psi\right) + m^2\psi + 2g\,\phi(x^\mu)\psi = 0. \tag{12}$$

To isolate the non-relativistic regime, we assume the induced metric is approximately flat, $g_{\mu\nu} \approx \eta_{\mu\nu}$, and restore physical units via

$$\Box_4 \to -\frac{1}{c^2}\frac{\partial^2}{\partial t^2} + \nabla^2, \quad m \to \frac{mc}{\hbar}, \quad \phi \to \frac{\phi}{\hbar},$$

where ∇^2 denotes the flat-space Laplacian on spatial slices. We define the slowly varying envelope field $\chi(t, \mathbf{x})$ by

$$\psi(t, \mathbf{x}) = e^{-imc^2 t/\hbar}\,\chi(t, \mathbf{x}), \tag{13}$$

and substitute into the equation of motion. Neglecting second-order time derivatives of χ, we obtain the Schrödinger equation

$$i\hbar\,\partial_t\chi = -\frac{\hbar^2}{2m}\,\nabla^2\chi + g\,\phi(t, \mathbf{x})\,\chi. \tag{14}$$

The effective potential

$$V_{\text{eff}}(t, \mathbf{x}) = g\,\phi(t, \mathbf{x}) \tag{15}$$

encodes all geometric modulation inherited from the higher-dimensional dynamics and recursive curvature. In this regime, quantum-like behavior such as interference and localization arises as the envelope response of χ to deterministic modulation [5, 7].

The transition from the full relativistic equation to the Schrödinger equation in Eq. (14) assumes that the matter field $\psi(x^\mu)$ evolves on timescales much slower than the carrier frequency $\omega = mc^2/\hbar$. This allows the ansatz $\psi(t, \vec{x}) = e^{-imc^2t/\hbar}\chi(t, \vec{x})$ to isolate rapid oscillations into the phase factor, leaving $\chi(t, \vec{x})$ as a slowly varying envelope.

Neglecting $\partial_t^2\chi$ is justified in the adiabatic regime, where

$$\left|\frac{\partial_t^2\chi}{\omega\,\partial_t\chi}\right| \ll 1,$$

and the energy contributions from kinetic corrections are subleading. Furthermore, the assumption of a weak, slowly varying geometric modulation $\phi(x^\mu)$ ensures that the effective potential $V_{\text{eff}} = g\phi$ does not introduce rapid oscillations that would break this hierarchy of scales.

This scale separation constitutes a Born–Oppenheimer-like limit, where the recursively modulated field ϕ plays the role of a classical background. Thus, this approximation is valid as long as backreaction remains weak and the system is near its ground state configuration.

4.6 Modulated Potential Structure and Interference

As shown in Section 3, the recursively evaluated field $\phi(x^\mu)$ inherits nonlinear structure from the curvature modulation of the recursive geometry. For an ambient plane wave modulated by rippled recursion functions, we have

$$\phi(x^\mu) = \cos\left(k_\mu x^\mu + \epsilon A \sin(\lambda k_\nu x^\nu + \theta)\right), \tag{16}$$

where $\epsilon \ll 1$ is a small geometric deformation amplitude, $A > 0$ is a composite modulation factor, λ is the modulation frequency, and θ is a phase offset. The resulting effective potential is

$$V_{\text{eff}}(x^\mu) = g \cos\left(k_\mu x^\mu + \epsilon A \sin(\lambda k_\nu x^\nu + \theta)\right). \tag{17}$$

In the weak modulation limit, a first-order expansion yields

$$V_{\text{eff}}(x^\mu) \approx g\cos(k_\mu x^\mu) - g\epsilon A \sin(k_\mu x^\mu)\sin(\lambda k_\nu x^\nu + \theta) + \mathcal{O}(\epsilon^2). \tag{18}$$

This describes a carrier oscillation modulated by a slowly varying envelope, producing beat-like interference patterns in the matter field amplitude [9].

4.7 Interpretation

The modulation induces regions of constructive and destructive interference in $\chi(t, \mathbf{x})$, leading to deterministic localization of the amplitude $|\chi|^2$. This behavior does not arise from stochastic collapse in this model, but from recursive phase resonance between ambient and confined wave components. The location and structure of interference nodes depend on the relative phase alignment between these fields, which is shaped by recursive geometry and dynamically governed by classical field behavior.

These results indicate that quantum-like interference, localization, and amplitude suppression can possibly emerge from classical field evolution in a self-consistently modulated recursive system. The Schrödinger equation, in this context, is a coarse-grained effective description of deeper recursive dynamics. The observable quantities are encoded in the recursively aligned field $\phi(x^\mu)$ and the matter envelope $\psi(x^\mu)$. We interpret $|\chi(t, \vec{x})|^2$, where $\psi = e^{-imc^2 t/\hbar}\chi$, as a coarse-grained measure of energy localization, not probability, in a deterministically modulated system.

Detection-like events in this classical model correspond to regions of constructive resonance—locations where the recursive modulation $\phi(x^\mu)$ aligns with $\psi(x^\mu)$, leading to transient energy localization. These nodal and anti-nodal structures mimic the probabilistic patterns observed in quantum detection experiments, but arise from deterministic geometric resonance.

While not fundamentally probabilistic, the envelope structure defines a functional analog to quantum probability density, enabling a possible reinterpretation of interference and localization as signatures of recursive field modulation, not stochastic measurement collapse [4].

5 Conservation Laws and Symmetry Considerations

We now address the question of conservation laws in the coupled system consisting of a real scalar field $\Phi(x^A)$ in the six-dimensional ambient spacetime $(\mathcal{M}_6, \eta_{AB})$, and a confined matter field $\psi(x^\mu)$ living on a recursively stabilized four-dimensional (as per the definition in this model) hypersurface Σ_4. The two fields are dynamically coupled. The confined field ψ acts as a localized source for Φ, while the induced return field $\phi(x^\mu)$ contributes an effective potential to the evolution of ψ, reflecting real energy exchange across a recursively modulated wave structure. Although dynamics within Σ_4 may appear non-conservative due to modulation, the full system remains conservative at the level of the higher-dimensional deterministic field theory.

The ambient field theory governing $\Phi(x^A)$ is invariant under global spacetime translations $x^A \to x^A + \epsilon^A$, from which Noether's theorem guarantees the existence of a conserved energy–momentum tensor [15, 21]. For the free field Φ, this tensor takes the standard form

$$T_\Phi^{AB} = \partial^A\Phi\,\partial^B\Phi - \frac{1}{2}\eta^{AB}\,\partial_C\Phi\,\partial^C\Phi, \tag{19}$$

satisfying $\partial_A T_\Phi^{AB} = 0$ in the absence of sources. However, when Φ is coupled to ψ through a curvature-localized source, the total energy–momentum tensor must be extended to include contributions from ψ and the interaction. The field $\psi(x^\mu)$, being restricted to the recursive layer Σ_4, enters the 6D energy budget through a delta-functional source term in the ambient equation $\Box_6\Phi = J(x^A)$, with J sharply localized across the recursive structure. The interaction term between Φ and ψ likewise contributes spatially modulated exchange terms to the total energy flow.

From the four-dimensional frame in this model, the apparent dynamics of ψ are governed by a Klein–Gordon-type equation with a time- and space-dependent potential determined by the recursively modulated field $\phi(x^\mu)$. The modulation observed in $\phi(x^\mu)$ arises from deterministic resonance between confined and ambient fields, shaped by recursive geometry, not by probabilistic effects. As a result, the effective four-dimensional energy–momentum tensor for ψ does not satisfy a naive conservation law, and instead obeys a divergence equation with a nonzero source term. Thus, this is not a violation of fundamental principles, but a structural feature of recursive systems, in which localized field layers exchange energy and momentum through coherent, deterministic coupling with higher-dimensional curvature modes.

Despite this apparent non-conservation in the 4D subsystem, the full six-dimensional structure remains conservative. The total energy–momentum tensor, which includes contributions from the ambient field Φ, the recursively confined field ψ, and their mutual coupling, satisfies the global conservation law $\partial_A T^{AB}_{\text{total}} = 0$. This ensures that all energy–momentum transfer on Σ_4 is accounted for by geometric flux into or out of the recursive structure. In this sense, apparent energy flow within the lower-dimensional field can be interpreted as the truncated view of recursive dynamics, analogous to dissipative terms in effective field theories derived from integrating out higher-dimensional coherence modes [22].

In special cases, the recursive geometry may preserve a residual symmetry. For instance, if the scalar phase function $S(x^\mu) = k_\mu x^\mu$ remains invariant under translation along a fixed direction in field space, then the system admits a partially conserved current. In the unmodulated limit, where the geometry is flat and the effective potential is translationally invariant, the current

$$J^\mu = \phi\,\partial^\mu\phi + \psi\,\partial^\mu\psi \tag{20}$$

satisfies $\partial_\mu J^\mu = 0$. However, when modulation is present through recursive curvature functions, this current acquires geometric source and sink terms, such that

$$\partial_\mu J^\mu = \mathcal{S}(x^\mu), \tag{21}$$

where the source function $\mathcal{S}(x^\mu)$ depends on derivatives of the recursive modulation functions and the field configurations. This term quantifies the local rate of energy–momentum exchange within the recursive coupling, and vanishes only in the limit of trivial geometry.

5.1 Interpretation

While the effective four-dimensional theory in this model may exhibit explicit breaking of translation invariance and corresponding non-conservation of energy and momentum, this behavior is a natural outcome of recursive energy exchange across dimensional strata. The full system remains globally conservative under six-dimensional Poincaré symmetry. Apparent non-conservation in 4D is a geometric manifestation of deterministic energy flow within a curvature-coupled recursive architecture. This structure supports the interpretation of interference, decoherence, and measurement-like features as emergent from deterministic resonance within a classical higher-dimensional system [8, 23].

6 Extension to Other Fields and Physical Realism

Although the present formulation focuses on a real scalar field $\Phi(x^A)$ propagating in the six-dimensional ambient space, a complete physical framework must incorporate all fundamental field types, including fermions, gauge bosons, and gravitational degrees of freedom. In this section, we briefly outline how the recursive modulation mechanism described here may extend to a more comprehensive and physically realistic setting [14, 21].

6.1 Fermionic Fields

A Dirac spinor field $\Psi(x^A)$ may be introduced in the ambient space, satisfying the six-dimensional Dirac equation:

$$(i\Gamma^A\partial_A - M)\Psi = 0,$$

where Γ^A are the appropriate six-dimensional gamma matrices. The observable fermionic field is then defined by recursive evaluation onto the curvature-modulated hypersurface:

$$\psi(x^\mu) = \Psi(x^\mu, f_1(x^\mu), f_2(x^\mu)).$$

To ensure consistency with local Lorentz invariance and spin structure on Σ_4, this construction requires the definition of an induced spinor bundle and a compatible frame field. The recursively modulated gamma matrices must transform appropriately under the recursive structure and may inherit resonance-induced modulation similar to that observed in the scalar sector. In such a model, interference, or decoherence-like phenomena, could arise from phase alignment across recursive curvature layers, not from curvature alone [23].

6.2 Gauge Fields

Gauge fields $A_A(x^B)$ in the ambient space, obeying standard Yang–Mills dynamics, may likewise be recursively evaluated onto Σ_4. The effective four-dimensional gauge potential is obtained via curvature-aligned tangents:

$$A_\mu(x^\mu) = A_A(x^B(x^\mu)) \frac{\partial x^A}{\partial x^\mu}.$$

This recursive coupling can encode effective variations in coupling constants, field strengths, or gauge phases, shaped by the resonance conditions defined by the recursive modulation. These variations arise from deterministic wave interaction across dimensional recursion layers, with curvature alignment acting as the structural mechanism, not passive embedding. As with scalar fields, geometric modulation may generate structured interference or amplitude suppression in gauge degrees of freedom, offering a deterministic analog to conventional quantum fluctuations [1].

6.3 Gravitational Degrees of Freedom

The recursive geometry of Σ_4 naturally gives rise to an induced metric $g_{\mu\nu}(x^\mu)$ and extrinsic curvature tensors $K^{(i)}_{\mu\nu}$. In our previous work (Surface Tension from Curved Hypersurfaces as the Origin of the Cosmological Constant), it was shown that effective gravitational dynamics, including surface tension, cosmological constant emergence, and response to curvature perturbations, can be derived directly from recursive curvature constraints. In the present formulation, we assume that the gravitational sector is already consistent with the recursive modulation framework. However, it is noteworthy that the same resonance mechanism may apply to curvature modes themselves, potentially giving rise to gravitational interference or phase-locked metric oscillations across recursive curvature strata.

6.4 Interpretation

Taken together, these extensions suggest that the scalar field resonance mechanism is not unique to bosons, but may reflect a general recursive principle governing how dimensional wave alignment generates structured field behavior. The key insight is that quantum-like features can emerge from deterministic resonance between field modes propagating in recursive dimensional systems and those stabilized within lower-order curvature layers.

In this framework, geometry is not merely a background or coordinate scaffold. Rather, it is an active resonant structure, shaping interference, localization, and amplitude modulation through recursive wave alignment. If so, this theory could provide a unified geometric origin for the entire spectrum of quantum-like phenomena—across bosonic, fermionic, and gauge-theoretic sectors [8, 19].

A complete treatment will require detailed development of spinor and gauge field recursion mechanics, and deeper exploration of how dimensional resonance, phase locking, and curvature filtering modulate all fundamental fields in analogy to the scalar case. These investigations are left for future work.

7 Geometric Modulation and Interference Without Superposition

Let the recursive geometry be defined by two smooth functions of a scalar phase $S(x^\mu) = k_\mu x^\mu$:

$$f_1(x^\mu) = \epsilon \sin(\lambda S), \qquad f_2(x^\mu) = \epsilon \cos(\lambda S),$$

where $\epsilon \ll 1$ sets the amplitude of hypersurface oscillations, and λ controls the modulation frequency. We consider a classical bulk wave in six-dimensional flat space of the form

$$\Phi(x^A) = \cos(k_\mu x^\mu + \omega_4 x^4 + \omega_5 x^5),$$

where k_μ is the four-dimensional wavevector, and ω_4, ω_5 describe propagation in the transverse directions. Evaluating this wave along the recursive layer yields the modulated field

$$\phi(x^\mu) = \cos\left(k_\mu x^\mu + \epsilon A \sin(\lambda k_\mu x^\mu + \theta)\right), \tag{22}$$

with $A = \sqrt{\omega_4^2 + \omega_5^2}$ and $\theta = \tan^{-1}(\omega_5/\omega_4)$.

This result describes a phase-modulated wave, exhibiting a classical carrier oscillation in $k_\mu x^\mu$, modulated by a slower envelope arising from deterministic resonance between confined and ambient field modes. The observed modulation reflects the phase structure introduced by wave activity across recursive curvature layers. The curvature serves as a static scaffolding for this interaction; it does not generate the interference, but provides structural alignment for phase resonance.

7.1 Comparison with Classical Interference

In conventional optics, interference arises from the coherent superposition of multiple wavefronts [6, 9]. A prototypical example is Young's double-slit experiment, where the intensity distribution is given by

$$I(x) \propto \cos^2(kx),$$

reflecting a fixed phase difference between two spatially separated sources.

In contrast, the present model contains only a single ambient wave. The observed structure emerges from the deterministic interaction of that wave with the recursive geometric modulation. The resulting envelope function

$$E(x) = \cos\left(\epsilon A \sin(\lambda k x + \theta)\right)$$

bounds the oscillations of the field $\phi(x^\mu)$ and generates spatial modulation patterns that visually resemble classical interference fringes. When squared, $E^2(x)$ mimics the intensity distribution of standard interference patterns, despite arising from deterministic recursion rather than probabilistic superposition.

This highlights a central principle of the model: that quantum-like interference may emerge not from amplitude uncertainty, but from structured resonance between dimensional wave fields coexisting in the same geometric framework. The recursive modulation guides this interaction spatially, but the interference itself arises from deterministic classical dynamics, not from probabilistic collapse or abstract superposition.

To illustrate this comparison visually, we plot the classical intensity profile $I(x) \propto \cos^2(kx)$ alongside the deterministic envelope function $E(x) = \cos(\epsilon A \sin(\lambda k x + \theta))$. The resulting overlay (see Figure 1) shows how the recursive modulation generates a structure closely resembling traditional interference fringes.

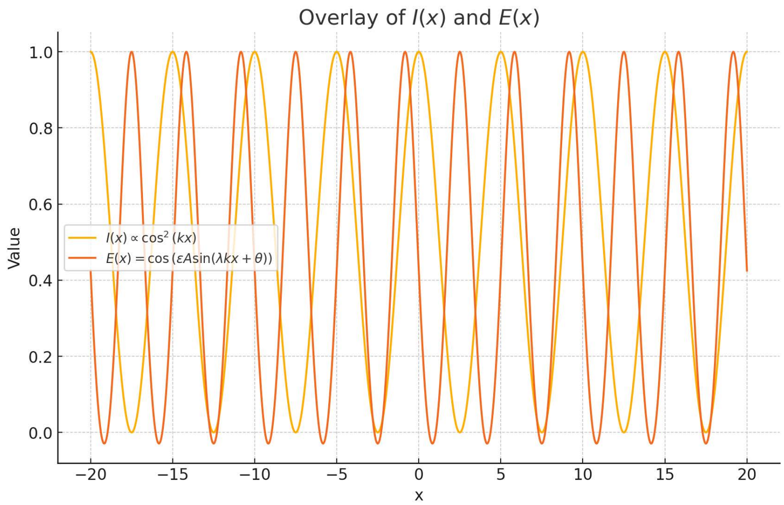

Fig. 1 Overlay of classical interference intensity $I(x) \propto \cos^2(kx)$ and recursive envelope function $E(x)$. Despite arising from different physical principles, both produce similar spatial modulation patterns.

7.2 Interpretation

These results appear to support the central thesis: that quantum-like interference phenomena can be understood not as fundamental superpositions, but as effective consequences of deterministic classical fields modulated by recursive geometry [4, 5]. The deterministic interaction between smooth ambient waves and a confined oscillating field within a shared recursive structure generates amplitude modulation, envelope suppression, and nodal interference, without invoking multiple trajectories, stochastic amplitudes, or observer collapse.

This points toward perhaps a new interpretation of quantum behavior. What appears as interference in spacetime may be the internal expression of phase resonance across recursively aligned wave structures. Curvature provides structural coherence to this process, but the interference pattern itself reflects deterministic classical field dynamics, not metric warping or probabilistic branching.

The model thus offers a fully classical, recursively geometric origin for phenomena traditionally attributed to quantum superposition.

8 Discussion

The framework proposed in this work draws on established ideas in higher-dimensional physics but advances them in a novel direction by explicitly demonstrating how quantum-like phenomena may arise from deterministic resonance between dimensional wave systems, structured through recursive geometry. While the mathematical structure of hypersurfaces in higher-dimensional manifolds has been previously explored, particularly in the work of Regge and Teitelboim [18] and in brane-world cosmology [16] [17], such studies have typically focused on gravitational dynamics or localization of classical fields, rather than the emergence of interference, quantization, or decoherence phenomena.

The closest conceptual precedents to this model can be found in the work of Jalalzadeh and Sepangi [10], who examined test particles confined to a 4D brane in a 5D bulk, and in subsequent extensions showing how quantization may result from periodic motion in the extra dimension [11]. However, those models treat quantum features as arising from discrete geodesics or classical brane fluctuations, rather than as continuous interference phenomena emerging from deterministic dimensional field alignment. In contrast, the present model derives explicit scalar field modulation caused by resonance across recursive curvature layers, and demonstrates that these modulations induce amplitude suppression, localization, and nodal behavior analogous to quantum measurement phenomena.

The work also aligns conceptually with deterministic approaches to quantum mechanics, including 't Hooft's cellular automaton framework [19, 20], and the Many-Interacting-Worlds (MIW) theory [8], which aim to recover quantum statistics from an underlying deterministic substrate. Unlike these theories, the present model does not require hidden variables, ensemble dynamics, or multiple ontological branches. Instead, it situates all dynamics within a single higher-dimensional classical field theory, where quantum-like behavior emerges from recursive modulation between bulk waveforms and confined field modes, structured by, but not caused by, recursive curvature.

From the perspective of decoherence theory, particularly Zurek's environment-induced framework [23], this model offers a geometric counterpart: coherence is suppressed not through probabilistic environmental averaging, but through resonance with unobserved higher-dimensional curvature modes. Recursive modulation replaces decoherence, without invoking stochasticity.

Furthermore, analog experiments such as the hydrodynamic pilot-wave systems developed by Couder and Fort [4] demonstrate that interference and localization phenomena typically attributed to quantum mechanics can emerge in classical systems under specific wave–particle coupling dynamics. The present model functions in a similar spirit, but with the key difference that the coupling arises naturally from recursive geometry, not an engineered medium. The resemblance between the deterministic envelopes derived in this theory and the observed interference structures in such systems supports the plausibility of classical wave resonance as a mechanism underlying quantum phenomena.

Taken together, these connections suggest that this model occupies a unique conceptual space, inheriting mathematical structure from differential geometry, physical intuition from analog experiments, and motivation from deterministic quantum reconstructions—while introducing a new mechanism: quantum-like behavior as the natural outcome of recursive field alignment across curvature-defined strata.

The interference patterns, amplitude suppression, and localization behavior derived here follow directly from geometric recursion. No superposition, no randomness, and no observer-driven collapse is present. As such, this framework can potentially offer a viable path toward reconciling quantum phenomenology with a deterministic classical theory grounded in recursive dimensional resonance.

While the present scalar formulation reproduces core quantum features, it does not claim to fully recover all quantum phenomena, such as Bell nonlocality or spin entanglement. These effects may require extension into spinor and gauge sectors, which we outline for future work. Nonetheless, the model establishes a classical geometric substrate capable of reproducing quantum-like statistics without postulated uncertainty.

9 Experimental Outlook

While the current formulation is highly theoretical, its core mechanism of modulation through recursive coupling of classical wave systems, lends itself to experimental investigation in analog platforms. We outline several concrete paths for testing the predictions of this model:

9.1 Optical Waveguide Analogs

A promising testbed lies in photonic lattices with engineered refractive index modulations. By designing a one-dimensional waveguide array with sinusoidally modulated coupling coefficients, recursive curvature can be effectively mimicked. The resulting field propagation should exhibit deterministic interference and envelope suppression analogous to Eq. (22), offering a direct optical analog of recursive modulation.

9.2 Hydrodynamic Pilot-Wave Experiments

The modulated scalar field solutions bear striking resemblance to the interference patterns observed in hydrodynamic pilot-wave systems developed by Couder and Fort [4]. These experiments have already demonstrated quantum-like statistics in macroscopic classical systems. A refined version of such experiments could be tailored to match the recursive modulation predicted by this model, offering a physical analog for recursive resonance without invoking quantum indeterminacy.

9.3 Numerical Simulation of Recursive Modulation

The modulated field

$$\varphi(x^\mu) = \cos\left(k_\mu x^\mu + \epsilon A \sin(\lambda k_\nu x^\nu + \theta)\right)$$

can be simulated numerically to exhibit spatial localization and nodal interference structures. By varying the modulation parameters ϵ, A, and λ, one can visualize the transition from smooth oscillation to envelope-suppressed, interference-like structures. This supports direct comparison with classical interference and provides a path toward synthetic validation.

9.4 Distinguishing Predictions

Unlike standard quantum mechanics, which predicts statistical distributions over many runs, this model predicts persistent, deterministic localization features under repeated initialization of the same field configuration. Observing consistent nodal suppression in analog systems could provide a signature distinguishing recursive resonance from probabilistic superposition.

9.5 Quantum Simulation Platforms

With the development of synthetic higher-dimensional quantum simulation platforms [**?**], it may soon be possible to encode recursive dimensional interactions in controlled quantum systems. Embedding this model's modulation dynamics into such frameworks would allow direct comparison between quantum interference and deterministic geometric resonance, enabling experimental falsifiability of the proposed mechanism.

10 Conclusion

This work presents a geometric mechanism by which quantum-like behavior can emerge from deterministic resonance between classical fields inhabiting recursively coupled dimensional layers. By modeling the observable universe as a three-dimensional hypersurface stabilized within a six-dimensional flat spacetime, we showed that structured interference patterns that are typically attributed to quantum superposition, can arise from phase alignment and modulation driven by field-on-field interaction across recursive curvature.

It is important to note that the observed modulation is not generated by curvature or projection alone, but by deterministic interaction between a confined field and a freely propagating higher-dimensional field. The recursive structure defines where and how this interaction occurs. The resulting amplitude suppression, localization, and nodal structures are not statistical artifacts. They, in principle and modeling, appear to be structural outcomes of classical wave resonance.

The model aims to perhaps offer a deterministic starting point to quantum interpretations, suggesting that what appears as interference in spacetime may in fact be the real-space signature of dimensional field interaction. These results may open a path toward reinterpreting quantum phenomena as emergent from recursive field behavior, not abstract probability, but structured, curvature-aligned classical resonance.

Declarations

- **Funding:** This work received no external funding.

- **Conflict of interest/Competing interests:** The author declares no competing financial or non-financial interests.
- **Ethics approval and consent to participate:** Not applicable.
- **Consent for publication:** Not applicable.
- **Data availability:** All data used are from publicly available sources, as cited in the references. No new datasets were generated.
- **Materials availability:** Not applicable.
- **Code availability:** No code was developed or used in this study.
- **Author contribution:** The author is solely responsible for the conception, derivation, analysis, writing, and editing of this manuscript.
- **Use of AI tools:** AI (ChatGPT-4o, OpenAI) was used under the author's direction to assist with mathematical derivations, symbolic computation, and conceptual clarification of physics topics. The author independently conceived the core ideas, selected and devised the appropriate models, interpreted all results, and wrote the paper. AI was used as an educational and computational aid; it did not generate the scientific content, arguments, or conclusions without author oversight.

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